

Mahanaz Syed, M.S., Ph.D.

Biomedical Informatics Researcher · Instructor, Population Health Sciences

UT Health San Antonio

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PROFILE

Biomedical informatics researcher with more than a decade of experience across the full span of healthcare data, from operational systems, business intelligence, and data warehousing to research informatics. My work centers on standardizing disparate clinical data and applying artificial intelligence, machine learning, and natural language processing to extract and harmonize high-value information from electronic health records. As an NIH/NCATS K12 Career Development Scholar, I lead research developing AI-enabled, standards-based tools for automated clinical fact extraction to advance interoperability and accelerate translational research.

RESEARCH INTERESTS

Artificial intelligence, machine learning, and natural language processing for healthcare data · Clinical data warehousing and integration of heterogeneous data sources · EHR data interoperability and standards (HL7 FHIR, OMOP CDM, PCORnet, i2b2) · Scalable informatics platforms for cancer and translational research · Real-world data · Health equity and representation in clinical research.

EDUCATION

Ph.D. in Biomedical Informatics	2021
<i>University of Arkansas for Medical Sciences, Little Rock, AR</i>	
• Mentored by Dr. Fred Prior and Dr. Meredith Zozus	
M.S. in Computer Engineering	2010
<i>University of Houston – Clear Lake, Houston, TX</i>	
B.Tech. in Electronics and Communication Engineering	2008
<i>Jawaharlal Nehru Technological University, Hyderabad, India</i>	

PROFESSIONAL EXPERIENCE

Instructor / Research, Population Health Sciences	2022 – Present
<i>UT Health San Antonio, San Antonio, TX</i>	
Research Data Analyst, Clinical Data Warehouse	2017 – 2022
<i>Department of Biomedical Informatics, University of Arkansas for Medical Sciences, Little Rock, AR</i>	
Applications Systems Analyst / Programmer — Senior, Business Intelligence	2016 – 2017
<i>University of Arkansas for Medical Sciences, Little Rock, AR</i>	
Programmer Analyst	2015 – 2016
<i>Iflowsoft Solutions Inc., Iselin, NJ</i>	
Applications Systems Analyst / Programmer, Business Intelligence	2011 – 2015
<i>University of Arkansas for Medical Sciences, Little Rock, AR</i>	
Programmer Analyst	2010 – 2011
<i>Orgspire Inc., Dublin, OH</i>	
Programmer Analyst	2010 – 2010
<i>Nth Talent Inc., Irving, TX</i>	

GRANTS AND FUNDED RESEARCH

NIH/NCATS K12 Career Development Award	2025 – 2027
• Harnessing AI and FHIR Standards for Automated Fact Extraction from EHR Data in Clinical Studies and Registries	
IIMS Pilot Grant, NIH/NCATS	2025 – 2026
• Unlocking Information in Legacy Medical Records for Research	
Site Principal Investigator — NIH/NIA, R01AG082698	2024 – 2029
• A deep learning algorithm to detect signs of cognitive impairment in electronic health records	
Site Principal Investigator — PCORI / Amgen, cvMOBIUS2	2023 – 2026
• CardioVascular Multicenter Observational Investigation of Lipid Care in the United States	
Site Principal Investigator — PCORI, PCORnet 3.0 Greater Plains Collaborative	2022 – 2024
Informaticist — NIH/NCI, U24CA215109	2017 – 2022
• TCIA Sustainment and Scalability: Platforms for Quantitative Imaging Informatics in Precision Medicine	

SELECTED PUBLICATIONS

Full list available on Google Scholar.

- Syed M, et al. Translating evidence into practice: adapting TrialGPT for real-world clinical trial eligibility screening. *Journal of the American Medical Informatics Association*, 2026.
- Li D, ... Syed M, et al. Clustering analysis of multi-site electronic health records reveals distinct subphenotypes in stage-1 acute kidney injury. *Communications Medicine*, 2025.
- Mazzotti DR, ... Syed M, et al. Positive Airway Pressure, Mortality, and Cardiovascular Risk in Older Adults With Sleep Apnea. *JAMA Network Open*, 2024.
- McDowell BD, ... Syed M, et al. Implementing Cancer Registry Data With the PCORnet Common Data Model: The Greater Plains Collaborative Experience. *JCO Clinical Cancer Informatics*, 2024.
- Garza MY, ... Syed M, et al. Comparing the Accuracy of Traditional vs. FHIR-Based Extraction of Electronic Health Record Data for Two Clinical Trials. *Studies in Health Technology and Informatics (MEDINFO)*, 2024.
- Syed M, et al. DeIDNER Model: A Neural Network Named Entity Recognition Model for Use in the De-identification of Clinical Notes. *BIOSTEC – HEALTHINF*, 2022.
- Syed M, et al. Application of Machine Learning in Intensive Care Unit (ICU) Settings Using MIMIC Dataset: Systematic Review. *Informatics*, 2021.
- Wang Z, ... Syed M, et al. Clinical Data Warehousing: A Scoping Review. *Journal of the Society for Clinical Data Management*, 2024.

HONORS AND APPOINTMENTS

IIMS Poster Award Recipient	2026
IIMS K12 Career Award Recipient	2025
IIMS Pilot Grant Recipient	2025
Conference Session Chair, Intl. Conference on Health Informatics	2022

TEACHING

TSCI 6203 — Practicum in Biomedical Data Science	2023 - Current
<i>UT Health San Antonio — Data science and translational research applications</i>	
BMIG 5012 — Introduction to Biomedical Informatics II	2018
<i>University of Arkansas for Medical Sciences — Database management systems in healthcare with SQL</i>	
Introduction to Informatics, College of Medicine (Pathology)	2019

INVITED TALKS AND PRESENTATIONS

Invited Panelist — IT Roundtable, Clinical Research Forum **2026**

Session: Informatics Innovations to Accelerate Clinical Research (virtual)

Application of Machine Learning in Genomics **2019**

AI Campus Demonstrations, Little Rock, AR

NoSQL: A Graph DB Hybrid Model for Analysis and Knowledge Discovery **2018**

Healthcare Data and Analytics Association (HDAA) Conference, Seattle, WA

Establishing an Enhanced Research Data Warehouse with Disease Registries **2018**

Healthcare Data and Analytics Association (HDAA) Conference, Seattle, WA

PROFESSIONAL SERVICE

Journal Peer Reviewer

- Nature; Journal of Medical Internet Research; Journal of Clinical and Translational Science; Journal of the American Medical Informatics Association Open; IEEE Access; AMIA; and others

Professional Memberships

- American Medical Informatics Association (AMIA); Healthcare Data and Analytics Association (HDAA); Health Level Seven International (HL7)

PROGRAMMING SKILLS

Programming Languages

- Python, R, SQL, JavaScript, C / C++

Methods and Algorithms

- Machine learning and deep learning algorithms; natural language processing and named entity recognition; advanced AI and large language models

Database Systems

- Microsoft SQL Server, PostgreSQL, MySQL, Snowflake

TECHNICAL CERTIFICATIONS

- Epic Clarity Ambulatory (CLR 205) · Epic Clarity Administrator (CLR 100) · Microsoft SQL Server Certified